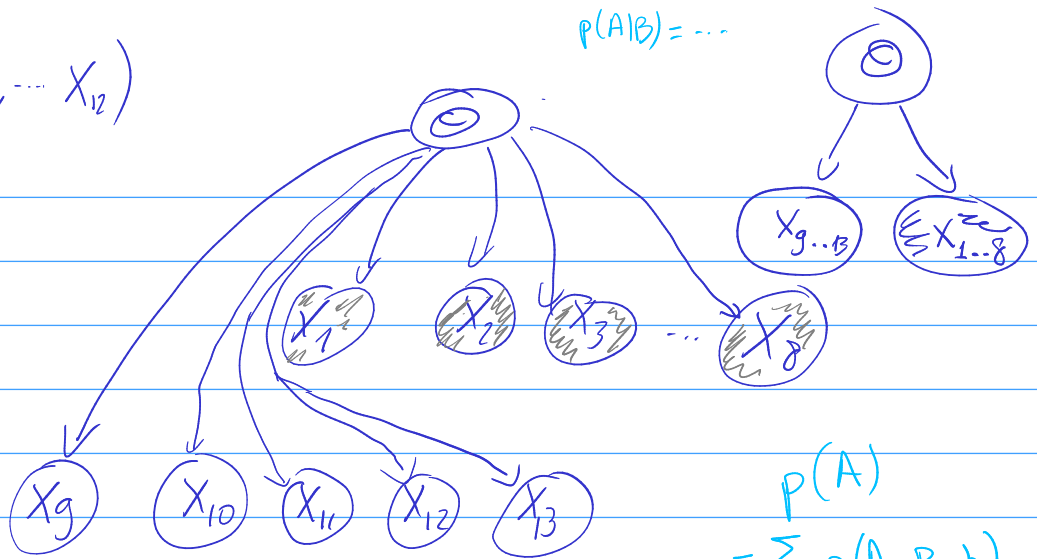


$$p(x_3 | \theta, x_1, x_2, x_4, x_5, \dots, x_{12})$$

$$= p(x_3 | \theta)$$

$$\forall i \quad p(x_i | \theta)$$



$$p(A)$$

$$= \sum_b p(A, B=b)$$

$$= \sum_b p(A|B=b) p(B=b)$$

$$p(x_9 = H, x_{10} = H, x_{11} = H, x_{12} = H, x_{13} = H \mid x_{1..8}) ?$$

$$= \underset{\substack{\text{prob. cond.} \\ \text{on } x_{1..8}}}{\uparrow} p_{x_{1..8}} \left(x_9 \dots x_{13} \right) = \int_0^1 p^{x_{1..8}}(x_9 \dots x_{13} \mid \theta) p^{y_{1..8}}(\theta) d\theta$$

$$= \int_0^1 p(x_9 \dots x_{13} \mid \theta, \cancel{x_{1..8}}) p(\theta \mid x_{1..8}) d\theta$$

$(x_i \perp x_j \mid \theta)$

bayes

$$\frac{p(x_{1..8} | \theta) p(\theta)}{p(x_{1..8})}$$

↑

↑
indep.

$$\frac{p(x_{1:n} | \theta) p(\theta)}{\int_0^1 p(x_{1:n} | \theta') p(\theta') d\theta'}$$